

SNP_GWAS: a flexible genome-wide association study software adapted for very large designs

T. Tribout¹, and D. Boichard^{1*}

¹Université Paris Saclay, INRAE, AgroParisTech, GABI, Jouy-en-Josas, France.

*Corresponding author: didier.boichard@inrae.fr

Introduction

Genome-wide association studies (GWAS) are the method of choice for detecting quantitative trait loci (QTL), mapping causal variants across the genome, and dissecting the genetic architecture of a complex trait. When performed at the whole genome sequence level (usually after imputation), this approach is expected to target a limited number of candidate variants. For a given variant effect, power and resolution are highly dependent on the size of the population analysed. In practice, only the largest QTLs are detected, which together explain only a small proportion of the total genetic variability in the trait. It is therefore crucial to analyse as large a population as possible. However, the model, even very simplified, must take into account the population structure to avoid potential confounding and false positives. Therefore, most approaches include a polygenic effect ($\mathbf{u}$) to account for the fact that individuals are related. The usual model includes the effects of the tested SNP, the polygenic effect, and other factors. Let us denote $\mathbf{G}$ the variance of the polygenic effect. If $\mathbf{G}$ is a matrix of genomic relationships built from SNP genotypes, this matrix is dense. Accordingly, if the number of individuals is large (e.g. more than 40,000), the computations become very long. Several solutions can be considered, such as replacing $\mathbf{G}$ by the pedigree relationship matrix $\mathbf{A}$, and $\mathbf{A}^{-1}$ is very sparse. Another solution is to set the smallest terms of $\mathbf{G}$ to zero, as in the fastGWAS option of GCTA (Yang et al, 2011) but this option is not adapted to most farm animal populations with many related individuals. The APY algorithm (Misztal et al, 2014) is also a way of reducing the size of the part of $\mathbf{G}$ to be inverted (Leite et al, 2024). Another solution proposed in this article is to replace $\mathbf{u}$ by $\mathbf{M}\mathbf{s}$, i.e. by the sum of the SNP effects multiplied by the centred genotypes. Both models are fully equivalent, as shown by Strandén and Garrick (2009). The advantage of this option is that the size of the $\mathbf{M}'\mathbf{M}$ matrix is equal to the number of SNPs and is constant regardless of the number of individuals analysed. In this article, we present a software that implements this option in an efficient way.

Several GWAS software packages already exist and are widely used (e.g., Yang et al, 2011). But four desirable features are not always proposed. (a) When phenotypes have different accuracies, weighing records by the inverse of their residual variance is desirable to increase power and reduce false positive rates. (b) When sequence variants are imputed from chip genotypes, their imputation accuracy is less than 1. Using dosages, i.e. expected number of alternative alleles, instead of genotypes is also desirable to maximise power and avoid biased estimates. (c) Dominance estimation is rarely proposed in addition to additive effect. (d) To maximise power, it is recommended to use the so-called LOCO procedure, i.e. to remove the markers of the chromosome of the tested SNP from the set of markers used to capture the family structure. However, because the LOCO procedure often generates inflation of statistical tests, the optimal method would be to exclude only markers in the region of the tested SNP, rather than the whole chromosome. All these features are included in the proposed software.

Methods

General approach. The general principle is to estimate, successively for each biallelic variant (called target variant (TV) below) considered, its additive, or additive and dominance effect(s) on the phenotype, using the following mixed model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}\mathbf{s} + \mathbf{Z}_k\mathbf{b}_k + \mathbf{e} \quad (1),$$

where $\mathbf{y}$ is a vector of n phenotypes, $\boldsymbol{\beta}$ is a vector of fixed effects including at least an overall mean, and possibly other fixed categorical or covariate effects, $\mathbf{s}$ is a vector of n_s SNP effects $N(\mathbf{0}, \mathbf{I} \sigma_s^2)$, with σ_s^2 being the genetic variance divided by $2 \sum_{j=1, n_s} p_j(1 - p_j)$ and p_j is the frequency of the second allele of the j^{th} SNP, $\mathbf{M}$ is an $n \times n_s$ matrix of centred genotypes with element $m_{ij} = g_{ij} - 2p_i$, where g_{ij} is the second allele dosage for the j^{th} SNP carried by the i^{th} individual, $\mathbf{b}_k$ includes the fixed additive effect, or the additive and dominance effects of the k^{th} TV, $\mathbf{Z}_k$ is the corresponding incidence matrix (details below), and $\mathbf{e}$ is the vector of residuals $N(\mathbf{0}, \mathbf{W}^{-1} \sigma_e^2)$, where $\mathbf{W}$ is a diagonal matrix which element i, w_{ii} , is the weight of i^{th} individual's phenotype y_i .

A typical situation is to use a set of 10,000 to 40,000 SNP to model genetic values and to test successively the effect of several hundred thousand to several million TV.

In the case where only the additive effect a_k of the k^{th} TV is estimated ($\mathbf{b}_k = a_k$), $\mathbf{Z}_k$ is a vector of either the discrete genotype or the predicted allelic dosage g_{ik} . When the dominance effect d_k is also estimated ($\mathbf{b}'_k = [a_k \ d_k]$), the entries in the 2^{nd} column of $\mathbf{Z}_k$ are equal to $|g_{ik} - 1| - 1$ if allelic dosages are considered.

Algorithm. The mixed model equations corresponding to the model in (1) are:

$$\begin{bmatrix} \mathbf{X}'\mathbf{W}\mathbf{X} & \mathbf{X}'\mathbf{W}\mathbf{M} & \mathbf{X}'\mathbf{W}\mathbf{Z}_k \\ \mathbf{M}'\mathbf{W}\mathbf{X} & \mathbf{M}'\mathbf{W}\mathbf{M} + k\mathbf{I} & \mathbf{M}'\mathbf{W}\mathbf{Z}_k \\ \mathbf{Z}'_k\mathbf{W}\mathbf{X} & \mathbf{Z}'_k\mathbf{W}\mathbf{M} & \mathbf{Z}'_k\mathbf{W}\mathbf{Z}_k \end{bmatrix} \begin{bmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{s}} \\ \hat{\mathbf{b}}_k \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{W}\mathbf{y} \\ \mathbf{M}'\mathbf{W}\mathbf{y} \\ \mathbf{Z}'_k\mathbf{W}\mathbf{y} \end{bmatrix} \quad (2)$$

with $k = \sigma_e^2 / \sigma_s^2$ assumed to be known.

In these equations, the $\mathbf{X}'\mathbf{W}\mathbf{X}$, $\mathbf{X}'\mathbf{W}\mathbf{M}$, $\mathbf{M}'\mathbf{W}\mathbf{M} + k\mathbf{I}$, $\mathbf{X}'\mathbf{W}\mathbf{y}$, and $\mathbf{M}'\mathbf{W}\mathbf{y}$ terms are constant for all studied TV. They are computed only once, and similarly the inverse $\mathbf{C}$ of the corresponding left-hand side $\begin{bmatrix} \mathbf{X}'\mathbf{W}\mathbf{X} & \mathbf{X}'\mathbf{W}\mathbf{M} \\ \mathbf{M}'\mathbf{W}\mathbf{X} & \mathbf{M}'\mathbf{W}\mathbf{M} + k\mathbf{I} \end{bmatrix}$ is computed once.

For each TV k , $\mathbf{b}_k$ is obtained after the absorption of $\boldsymbol{\beta}$ and $\mathbf{s}$ equations:

$$\left(\mathbf{Z}'_k\mathbf{W}\mathbf{Z}_k - [\mathbf{Z}'_k\mathbf{W}\mathbf{X} \ \mathbf{Z}'_k\mathbf{W}\mathbf{M}] \mathbf{C} \begin{bmatrix} \mathbf{X}'\mathbf{W}\mathbf{Z}_k \\ \mathbf{M}'\mathbf{W}\mathbf{Z}_k \end{bmatrix} \right) \hat{\mathbf{b}}_k = \mathbf{Z}'_k\mathbf{W}\mathbf{y} - [\mathbf{Z}'_k\mathbf{W}\mathbf{X} \ \mathbf{Z}'_k\mathbf{W}\mathbf{M}] \mathbf{C} \begin{bmatrix} \mathbf{X}'\mathbf{W}\mathbf{y} \\ \mathbf{M}'\mathbf{W}\mathbf{y} \end{bmatrix} \quad (3)$$

also denoted as

$$\mathbf{LHS}_k^{-1} \hat{\mathbf{b}}_k = \mathbf{RHS}_k \quad (4)$$

Note that solving this equation is straightforward since the size of this system is 1 or 2. This absorption approach also provides the error variance matrix of $\hat{\mathbf{b}}_k$, $\mathbf{EV}_k = \mathbf{LHS}_k^{-1} \hat{\sigma}_{\text{res}}^2$

The residual variance $\hat{\sigma}_{\text{res}_k}^2$ for the k^{th} TV can be calculated in a quick, approximate way ($\hat{\sigma}_{\text{res}_{k\text{ap}}}^2$), or in an exact but more time-consuming way ($\hat{\sigma}_{\text{res}_{k\text{ex}}}^2$):

In the approximate calculation, the $\boldsymbol{\beta}$ and $\mathbf{s}$ effects are considered independent of the TV effect ($\mathbf{b}_k$) and are calculated once for all by solving the system deprived of the $\mathbf{b}_k$ equations:

$$\begin{bmatrix} \hat{\boldsymbol{\beta}}_{\text{ap}} \\ \hat{\mathbf{s}}_{\text{ap}} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{W}\mathbf{X} & \mathbf{X}'\mathbf{W}\mathbf{M} \\ \mathbf{M}'\mathbf{W}\mathbf{X} & \mathbf{M}'\mathbf{W}\mathbf{M} + \rho\mathbf{I} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{X}'\mathbf{W}\mathbf{y} \\ \mathbf{M}'\mathbf{W}\mathbf{y} \end{bmatrix}$$

$$\text{and then } \hat{\sigma}_{\text{res}_{k\text{ap}}}^2 = (\mathbf{y}'\mathbf{W}\mathbf{y} - \hat{\boldsymbol{\beta}}'_{\text{ap}}\mathbf{X}'\mathbf{W}\mathbf{y} - \hat{\mathbf{s}}'_{\text{ap}}\mathbf{M}'\mathbf{W}\mathbf{y} - \hat{\mathbf{b}}'_k\mathbf{Z}'_k\mathbf{W}\mathbf{y}) / (n_s - r) \quad (7)$$

where r is the rank of the system in Eq. (2).

In the exact calculation, the actual β and s estimates are calculated for the k^{th} TV by adjusting the phenotypes y for $\hat{b}_k$ estimates as $y_{\text{adj}_k} = y - Z_k \hat{b}_k$, solving:

$$\begin{bmatrix} \hat{\beta}_{\text{ex}} \\ \hat{s}_{\text{ex}} \end{bmatrix} = \begin{bmatrix} X'WX & X'WM \\ M'WX & M'WM + kI \end{bmatrix}^{-1} \begin{bmatrix} X'Wy_{\text{adj}_k} \\ M'Wy_{\text{adj}_k} \end{bmatrix},$$

and then computing $\hat{\sigma}_{\text{res}_k}^2 = (y'Wy - \hat{\beta}_{\text{ex}}'X'Wy - \hat{s}_{\text{ex}}'M'Wy - \hat{b}_k'Z_k'Wy)/(n_s - r)$ (8)

The statistic associated with the test of the null hypothesis for the additive (and dominance if included in the model) effect is then:

$$T_{a_k} = \hat{a}_k / \sqrt{\text{LHS}_k^{-1}[1; 1] \hat{\sigma}_{\text{res}_k}^2} \quad \text{and} \quad T_{d_k} = \hat{d}_k / \sqrt{\text{LHS}_k^{-1}[2; 2] \hat{\sigma}_{\text{res}_k}^2} \quad (9)$$

LOSO option (Leave One Segment Out). This option is similar to the usual LOCO option but can be limited to a small region surrounding the TV. Let us assume that n_k markers around the tested variants are removed from the set of SNP used to account for the population structure. Instead of a complete reconstruction of C for each TV (which would be computationally prohibitive), the program computes the reduced C according to partitioned matrix rules:

if $C = \begin{bmatrix} C^k & C^{k\bar{k}} \\ C^{\bar{k}k} & C^{\bar{k}\bar{k}} \end{bmatrix}$, then $C_k = C^k - C^{k\bar{k}} (C^{\bar{k}\bar{k}})^{-1} C^{\bar{k}k}$.

Note that $C^{\bar{k}\bar{k}}$ corresponds to the marker to remove. It is a $\bar{n}_k \times \bar{n}_k$ matrix much smaller than C and can be easily and quickly inverted. In this way, C_k is obtained at low computing cost and can be used in the different equations above.

Implementation

The software tests the effect of variants chromosome by chromosome, by parallelizing analyses of groups of variants carried by a same chromosome. The software developed in Fortran 90 makes extensive use of optimized matrix calculation functions from the MKL library. The GWAS are carried out in 4 successive steps using the same software, which can easily be chained together using a shell script. The software requires a parameter file, a phenotype file, a genotype file with the SNP used to model the genetic value and the corresponding map file, the file of the genotypes or dosages of the TV and the corresponding map file. The parameter file provides the name of the files, the genetic and residual variances of the trait, the effects of the GWAS model, as well as the various options to be applied.

Step 1. This step builds the C matrix and the right-hand side. It is common to all chromosomes and performed only once. The file of the n_s SNPs used for population structure can be provided by the user, or these SNP can be selected by the program among the whole genome variants, optimizing spacing and MAF. C and right-hand side are stored in binary format.

Step 2. The second step is executed for each chromosome carrying TVs. These TVs are split into subgroups, the size of which is set by the user. For each subgroup, a parameter file and a binary file containing the genotypes/allelic dosages of the variants in the subgroup are created. In the latter file, genotypes are stored with one line per TV, allowing the step 3 to read this file line by line and minimize memory requirement.

Step 3. This is GWAS stricto sensu. The computations in each subgroup on each chromosome being independent, they can be run simultaneously on different processors in a cluster. For each subgroup, the program reads the binary files containing C and **RHS** created in step 1. If the user wishes to eliminate markers from a segment of given length surrounding the variants to be tested (or the whole chromosome), the C_k matrix is obtained from matrix C . Due to the way the variant subgroups were formed, C and **RHS**, or C_k and **RHS_k**, to be used for the subsequent calculations are identical for all variants in the subgroup. Then, for each TV, the program successively reads the Z'_k line in the binary file created in step 2, and calculates $\hat{b}_k$ as shown in Eq. (5). Regarding the error variance EV_k of $\hat{b}_k$, a convenient choice is to compute

the approximate one for all TV, and the exact one only when the approximate one exceeds a given $-\log(p)$ threshold provided by the used (eg, 3). This option saves computations while providing exact results for the significant variants. Both tests are provided in the output. A result file is created for each subgroup, containing for each variant its identification, location, $\hat{a}_k$, $T_{\hat{a}_k}$, $Pval_{\hat{a}_k}$ values, as well as $\hat{d}_k$, $T_{\hat{d}_k}$, $Pval_{\hat{d}_k}$ values if dominance effect was estimated.

Step 4. It aggregates the results of subgroups in a single file per chromosome.

Numerical Example

The real dataset used as an example corresponded to age at first calving in Montbéliarde dairy cattle breed, a trait with heritability equal to 0.06. Only chromosome 21 is presented. The dataset included 145,705 genotyped cows with phenotype. A set of 13,826 SNPs were selected from SNP-array genotypes on the 29 autosomes to adjust for population structure. One SNP was selected in each 160kb bin over the genome, and only SNPs with a $MAF > 0.10$ were retained. Sequence variants were imputed in a two-step procedure, from 50k to HD with FImpute using a reference of 526 HD genotyped ancestors, and from HD to sequence level in 8 batches with Minimac4, with a multibreed reference of 2712 bulls. In total, 959,762 variants were imputed and dosages were analysed in GWAS. Analyses were split in 134 jobs in parallel, with one analysis per subgroup of 7500 TVs. The SNP on a 10 Mb segment surrounding the TV were excluded. Exact tests were computed when the approximate $-2\log_{10}(p\text{-value})$ was greater than 2. Memory requirements were low (4 Gb/job), enabling massively parallel computing. Program optimisation is in progress to reduce the currently quite lengthy computational time. GWAS results are presented in Figure 1.

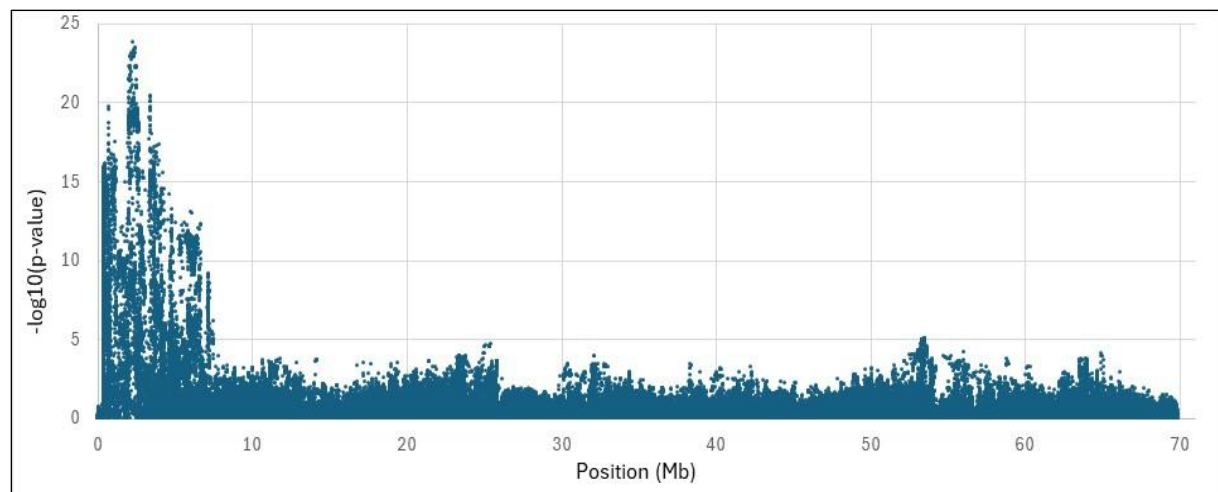


Figure 1. Manhattan plot of the GWAS analysis of age at calving on chromosome 21.

Conclusion. SNP_GWAS can cope with very large populations and provides options which are rarely proposed in existing software. It is publicly available at the following git link: https://github.com/thierrytribout/SNP_GWAS

References

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